

DATA ANALYSIS

Table 1: Volumes and Density Regular 3D Shapes

	Volume Equation	Calculated Volume (mm ³)	Calculated Volume (cm ³)	Calculated Density (g/cm ³)	Deviation (g/cm ³) $ \bar{(\bar{I}} - X_i) $
(1) Cube	$a \cdot b \cdot c$	2.584×10^4	2.584×10^{-1}	6.144×10^{-1}	2.43×10^{-2}
(2) Cuboid	$a \cdot b \cdot c$	5.136×10^4	5.136×10^{-1}	6.110×10^{-1}	2.77×10^{-2}
(3) Triangular Prism	$\frac{1}{2} \cdot a \cdot b \cdot c$	2.487×10^4	2.487×10^{-1}	5.497×10^{-1}	8.90×10^{-2}
(4) Cylinder	$\frac{1}{4} \pi \cdot a \cdot b \cdot c$	4.028×10^4	4.028×10^{-1}	5.276×10^{-1}	1.111×10^{-1}
(5) Square Pyramid	$\frac{1}{3} \cdot a \cdot b \cdot c$	1.669×10^4	1.669×10^{-1}	7.001×10^{-1}	6.14×10^{-2}
(6) Triangular Pyramid	$\frac{1}{6} \cdot a \cdot b \cdot c$	7.429×10^3	7.429×10^{-2}	5.982×10^{-1}	4.05×10^{-2}
(7) Cone	$\frac{1}{12} \pi \cdot a \cdot b \cdot c$	1.233×10^4	1.233×10^{-1}	8.179×10^{-1}	1.792×10^{-1}
(8) Ovoid					
(9) Ellipsoid	$\frac{1}{6} \pi \cdot a \cdot b \cdot c$	2.500×10^4	2.500×10^{-1}	6.900×10^{-1}	5.13×10^{-2}
(10) Sphere	$\frac{1}{6} \pi \cdot a \cdot b \cdot c$	1.371×10^4	1.371×10^{-1}	6.395×10^{-1}	8×10^{-4}

Using of the Densities of the Regular 3D Shapes, Calculate the Mean (Average) Density of Beech Wood

 6.387×10^{-1}

Using the Mean Density of Beech Wood and the Mass of Your Ovoid, Calculate the Volume of the Ovoid

 2.532×10^{-1}

Calculate the Mean (Average) Deviation of the Density of Beech Wood

 6.503×10^{-2}

Calculate the Relative Deviation (in %)

10.18 %

Calculate the Percent Error (in %)

10.0 %

Calculate the Percent Error (in ppt)

100 ppt

Table 2: Measurements of Fluorite

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	Calculated Density (g/mL)	Deviation (g/cm ³) $ (\bar{X} - X_i) $		Calculated Density (g/mL)	Deviation (g/cm ³) $ (\bar{X} - X_i) $
1	3.0×10^0	3.5×10^{-1}	11	3.1×10^0	2.5×10^{-1}
2	3.1×10^0	2.5×10^{-1}	12	3.2×10^0	1.5×10^{-1}
3	4.0×10^0	6.5×10^{-1}	13	2.4×10^0	9.5×10^{-1}
4	3.5×10^0	1.5×10^{-1}	14	3.7×10^0	3.5×10^{-1}
5	3.0×10^0	3.5×10^{-1}	15	2.5×10^0	8.5×10^{-1}
6	3.7×10^0	3.5×10^{-1}	16	3.3×10^0	5×10^{-2}
7	4.1×10^0	7.5×10^{-1}	17	2.7×10^0	6.5×10^{-1}
8	3.3×10^0	5×10^{-2}	18	3.8×10^0	4.5×10^{-1}
9	4.2×10^0	8.5×10^{-1}	19	3.2×10^0	1.5×10^{-1}
10	3.3×10^0	5×10^{-2}	20	2.9×10^0	4.5×10^{-1}

Using of the Densities of Fluorite, Calculate the Mean (Average) Density of Fluorite

 3.35×10^0

Calculate the Mean (Average) Deviation of the Density of Fluorite

 4.05×10^{-1}

Calculate the Relative Deviation (in %)

12.1 %

Calculate the Percent Error (in %)

 $5.21\% < \text{Error} < 5.51\%$ *(As density ranges)*

Calculate the Percent Error (in ppt)

 $52.1\text{ppt} < \text{Error} < 55.1\text{ppt}$ *(As density ranges)*